

A Tutorial:- 06

Q-1 Explain data constraints with mesh & hyper cube.

1) Data concentration with mesh.

In a p-processor interconnection network assume that there are d data items distributed arbitrarily with at most one data item per processor. The problem of data concentration is to move the data into first d processor of the network one data item per processor.

This problem is also known as packing. on a mesh, we might require the data items to move according to any indexing scheme of our choice. for example, the data could be moved into the first $\left\lceil \frac{d}{p} \right\rceil$ rows.

Let L be a p-processor linear array with d data item. To find the destination of each data item, we make use of a variable x_i . If processor i has a data item, then it sets $x_i = 1$; otherwise it sets $x_i = 0$. Let the prefix of the sequence $x_1, x_2, \dots, x_p$ be $y_1, y_2, \dots, y_p$. If processor i has a data item, then the destination of this item is y_i . The destination for the item is having been determining they are routed.

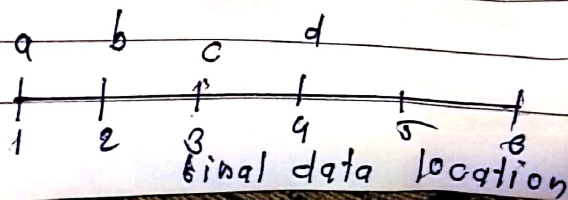
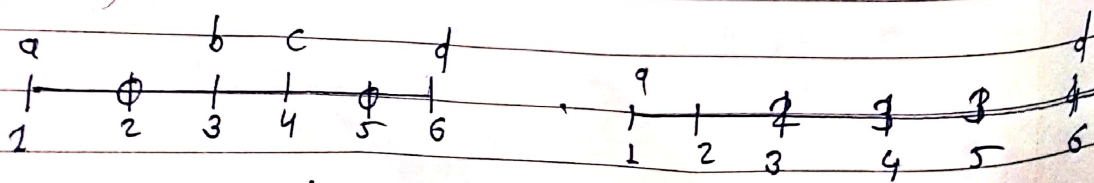


fig Data Concentration on a linear array

prefix computation as well as packed routing on a linear array takes p time steps each. then the total run time is $2p$.

Data Concentration with hyper cube.

on Hd assume that there are $k \times p$ data items distributed arbitrarily with at most one datum per processor
 the problem of data concentration is $0, 1, \dots, k-1$ of Hd one data item per processor

property 1: If the level d processors & incident links are eliminated from B_d two copies of B_{d-1} result. An example shows in fig. removal of level 3 preprocessor & links results in two independent B_2 's one of the have butterfly only even rows & the other consists only odd rows.

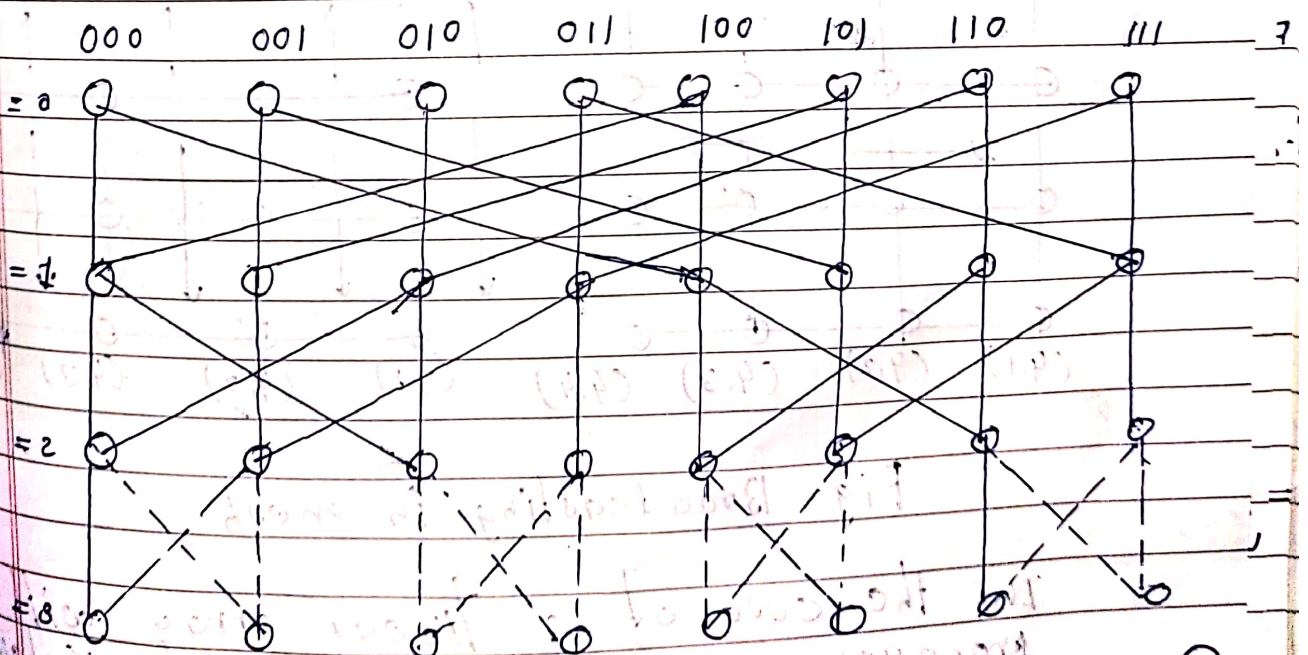


fig Removal of level d processors &

properly 2 :- All processors at level d are connected by a full binary tree. For ex, if we trace all the descendants of the processor $00 \dots 0$ of level 2000, the result is a full binary tree with the processor of level d as its leaves. In fact this is true for each processor at zero.

Q.2 : Explain prefix sum computation with the help of mesh.

→ Let E be any domain in which the binary associative unit time computable operator $\oplus$ is defined. Recall the prefix computation problem on E has an input n elements from E , say $x_1, x_2, \dots, x_n$. The problem is to compute the n elements $x_1, x_1 \oplus x_2, x_1 \oplus x_2 \oplus x_3, \dots, x_1 \oplus x_2 \oplus \dots \oplus x_n$.

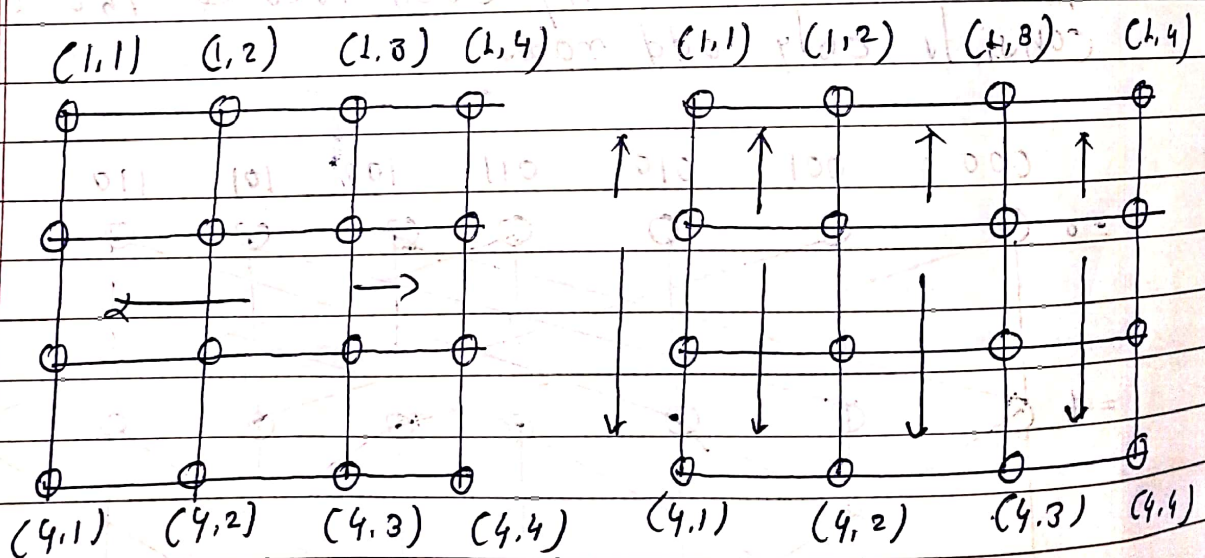


Fig Broadcasting in mesh

In the case of a linear array with processors assume that there is an element x_i at processor i , we have to compute

the prefixes $\alpha_1, \alpha_2, \alpha_p$
 After this computation processor this processor
 or i should have the value $\sum_{j=1}^p \alpha_j$

0	1	4	2
1	0	2	1
1	0	0	2
0	1	2	8

a]

0	1	2	4
1	1	3	4
1	1	1	3
0	1	2	5

b]

0	1	2	4	4
1	1	3	4	8
1	1	1	3	11
0	1	3	6	17

0	1	2	4
5	5	7	8
9	9	9	11
11	12	14	17

c]

d]

Fig prefix computation

phase 1: Row i computes the prefix of
 its J_p element. At the end the processor
 (i, j) has $y(i, j) = \sum_{q=1}^j \alpha_q$

phase 2: only column J_p computes prefix
 of sums computed in phase 1. Thus at the
 end processor (i, J_p) has $z(i, J_p) = \sum_{q=1}^p y(i, q) =$
 $\sum_{q=1}^p \sum_{j=1}^q \alpha_j$

phase 3: Broadcast $z(i, J_p)$ in row i .
 Node j in row i finally updates its result
 to $z(i, J_p) \oplus y(i+1, j)$

Q.3 Explain hyper cube Computational model.
→ A hypercube of dimension d , denoted H_d , has $p = 2^d$ processors. Each processor in H_d can be labeled with a d -bit binary number. For example the processor of H_3 can be labeled with 000, 001, 010, 011, 100, 101, 110 & 111.

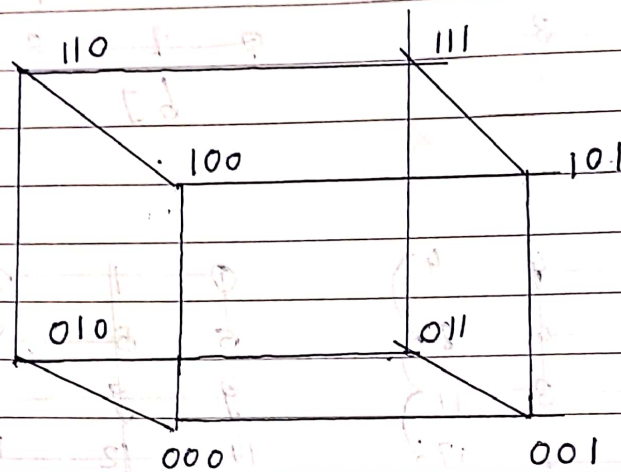


Fig A three-dimensional hypercube

We use the same symbol to denote a processor & its label, that is v is a d -bit binary number, then the first bit of v is the most significant bit of v . The second bit of v is the next-most significant bit of v . The second bit of v is the next-most significant bit. so on. the d th bit of v is its least significant bit.

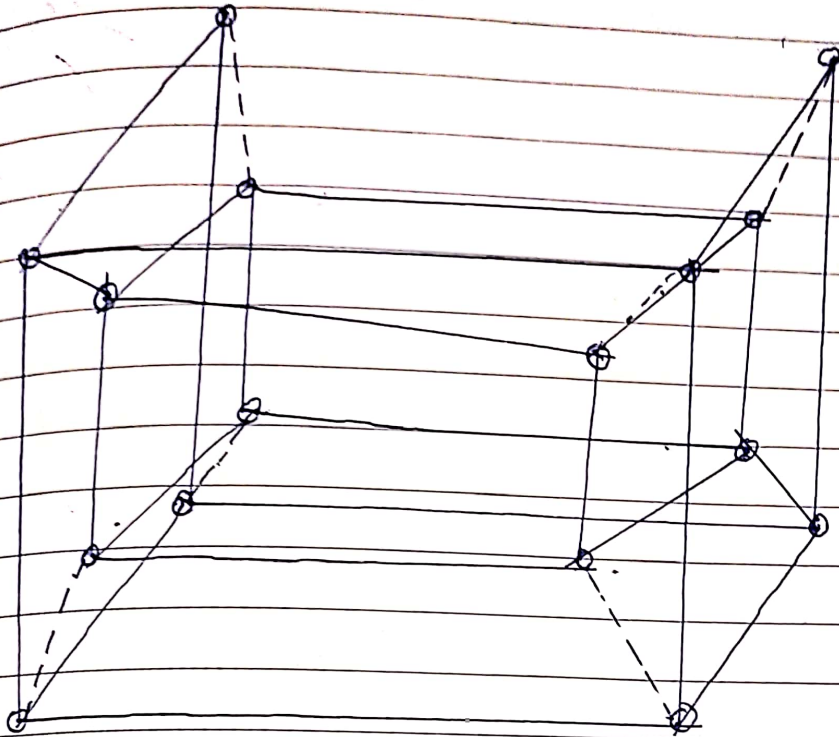


fig A hypercube of dimension four.

A hypercube network possesses numerous special features. One is its low diameter. If there are p processors in a hypercube then its diameter is only $\log p$. On the other hand, a mesh with the same number of processors has a diameter of $2(\sqrt{p} - 1)$. Also, hypercube H_{d+1} can be built recursively as follows: Take two identical copies of H_d , called them H' & H'' . If u is any processor of H' , connect it with its corresponding processor in H'' .